A Markov Decision Process Formulation of the Continuous Art Gallery Problem

Document 1 of 2: States, Actions, Dynamics and Reward

AGP Project — Design Note

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Abstract

We give a precise Markov Decision Process (MDP) formulation of the continuous Art Gallery Problem (AGP) with holes, as the decision problem solved by the PPO agent in `Polygons dataset/ppo_agp.py`. The formulation is deliberately built so that a *staged* training curriculum — first saturate the gallery with guards, then prune them — is expressible purely as a reweighting of a single fixed reward basis, with no change to the state space, the action space or the transition kernel. Two structural results organise the design. First, the coverage set function is monotone submodular, so the *exclusive area* of a guard (what is lost if it is removed) can only shrink as more guards are added; redundancy is therefore a quantity that only becomes measurable *after* saturation, which is the formal justification for the saturate-then-prune ordering. Second, both dense reward components we use are potential-based in the sense of Ng et al., and their discounted sum telescopes *exactly* into the sparse terminal objective "cover the gallery with as few guards as possible". Consequently the dense signal introduces no bias, and the curriculum controls only *when* a gradient exists, never *what* is optimal. Auxiliary terms that are not potential-based are identified explicitly, and we adopt the rule that precisely those terms must be annealed to zero. The companion document specifies the three-stage PPO curriculum built on top of this MDP.

Contents

1 Scope and relation to the existing code **2**
- 1.1 The problem . . . 2
- 1.2 Why write the MDP down . . . 2
- 1.3 The two documents . . . 3

2 Geometric preliminaries **3**
- 2.1 Gallery instances . . . 3
- 2.2 Visibility and coverage . . . 3
- 2.3 Submodularity, and why pruning must come second . . . 3

3 The decision process **4**
- 3.1 A contextual family of deterministic MDPs . . . 4
- 3.2 State . . . 5
- 3.3 Action space . . . 5
- 3.4 Transition . . . 6
- 3.5 Initial state and instance distribution . . . 6
- 3.6 Termination and horizon . . . 6
- 3.7 Discount . . . 7

4 Observation: a POMDP with a privileged critic **7**

5 Reward **8**
5.1 The objective we actually want . . . 8
5.2 Two potentials . . . 8
5.3 Which terms must be annealed . . . 9
5.4 The reward basis . . . 10
5.5 Weight calibration . . . 11
5.6 Reward groups and the multi-critic split . . . 12

6 Constraints **12**

7 The environment step, in full **13**

8 Mapping onto the current implementation **14**

9 Evaluation metrics **15**

10 Summary **15**

1 Scope and relation to the existing code

1.1 The problem

An *art gallery* is a simple polygon, possibly with polygonal holes. A *guard* is a point inside the gallery; it sees every point connected to it by a segment lying entirely inside the gallery. The classical AGP asks for the smallest set of guards whose union of visibility regions is the whole gallery. Chvátal's theorem [3], with Fisk's proof [7], gives the tight worst-case bound of $\lfloor n/3 \rfloor$ vertex guards for a simple n-gon, but finding a *minimum* guard set is NP-hard [11], hard to approximate below a logarithmic factor [5], and in the continuous (point-guard) setting the decision problem is $\exists\mathbb{R}$-complete [1]. Exact solvers exist but scale poorly and are usually restricted to vertex or grid guard candidates [4, 19]. This combination — geometric, combinatorial, NP-hard, cheap to simulate, with a natural incremental decision structure — is exactly the profile that makes a learned construction heuristic attractive [10, 13].

1.2 Why write the MDP down

The current implementation already *is* an MDP, but it was assembled incrementally: reward terms were added to fix specific failure modes (aiming outside the polygon, stacking guards, deterministic-evaluation deadlock) rather than derived from a stated objective. Writing the MDP down serves three purposes.

(i) It separates the *objective* (what we want) from the *shaping* (how we make it learnable), which in turn tells us which reward terms may stay fixed forever and which must be annealed away.

(ii) It makes the staged curriculum a property of the *weights*, not of the environment, so a single environment implementation serves all stages. Every locomotion paper we draw on does exactly this: Wang et al. [21] and Xu et al. [22] both keep one reward basis and publish one weight column per stage.

(iii) It exposes the action space that the guard-removal idea requires. Pruning is not expressible in the current $\mathcal{A} = [0, 1]^2$ "place a guard" action space at all, so the extension has to be stated before it can be trained.

1.3 The two documents

This document defines $\mathfrak{M}$: state, action, dynamics, observation, reward basis and constraints, together with the calibration of the reward weights. The companion document, *A Three-Stage PPO Curriculum for the Art Gallery Problem*, fixes the weight schedule per stage, the stage-transition criteria, the network architecture and the PPO hyperparameters, and gives the evaluation and ablation protocol.

2 Geometric preliminaries

2.1 Gallery instances

Definition 1 (Gallery)**.** A *gallery instance* is a pair $G = (P, \mathcal{H})$ where $P \subset \mathbb{R}^2$ is a simple polygon with vertex list $(p_1, \dots, p_n)$ and $\mathcal{H} = \{H_1, \dots, H_m\}$ is a finite set of pairwise disjoint simple polygons with $H_i \subset \operatorname{int} P$. The *free space* is the closed region

$$\mathcal{F}(G) \;=\; P \setminus \bigcup_{i=1}^{m} \operatorname{int} H_i, \qquad A \;=\; |\mathcal{F}| \;>\; 0,$$

with $|\cdot|$ denoting Lebesgue measure in the plane. We write $\partial\mathcal{F}$ for its boundary (outer boundary together with all hole boundaries) and $d_\partial(q) = \operatorname{dist}(q, \partial\mathcal{F})$.

In the repository an instance is a JSON record with keys `outer_points`, `holes`, `area` and `difficulty`, produced by `generate_polygon_dataset.py` in four difficulty tiers (`easy`, `mid`, `hard`, `extreme`) that scale the vertex count (8–12 up to 24–40), the irregularity and spikiness of the outer boundary, and the number of holes (1–3 up to 8–14). The tiers will serve as the instance-difficulty curriculum in Document 2.

2.2 Visibility and coverage

Definition 2 (Visibility polygon)**.** For $g \in \mathcal{F}$, the visibility polygon is

$$\mathcal{V}(g) \;=\; \{\, q \in \mathcal{F} \;:\; \overline{gq} \subseteq \mathcal{F} \,\}.$$

`Security.get_area_coverage_of_guard` computes $\mathcal{V}(g)$ exactly by the standard angular ray-sweep: rays are cast at every polygon and hole vertex angle, perturbed by $\pm\varepsilon$ to capture the two sides of each occluding reflex vertex, the nearest boundary hit along each ray is retained, and the hits are assembled in angular order and clipped back to $\mathcal{F}$. The result is an exact polygon, not a raster approximation — this matters, because it means every area quantity below is computed in continuous geometry even though the agent *perceives* the gallery as an image.

Definition 3 (Coverage set function)**.** For a finite $S \subset \mathcal{F}$ let

$$\mathcal{C}(S) \;=\; \bigcup_{g \in S} \mathcal{V}(g), \qquad f(S) \;=\; \frac{|\mathcal{C}(S)|}{A} \in [0,1], \qquad \mathcal{U}(S) \;=\; \mathcal{F} \setminus \mathcal{C}(S).$$

We call $f(S)$ the *coverage fraction* and $u(S) = 1 - f(S)$ the *uncovered fraction*.

2.3 Submodularity, and why pruning must come second

Proposition 4 (Monotone submodularity)**.** *f is normalised ($f(\emptyset) = 0$), monotone ($S \subseteq S' \Rightarrow f(S) \le f(S')$) and submodular: for all $S \subseteq S' \subset \mathcal{F}$ and all $g \in \mathcal{F}$,*

$$f(S \cup \{g\}) - f(S) \;\ge\; f(S' \cup \{g\}) - f(S').$$

Proof. f is a (continuous) coverage function: $f(S) = \nu\big(\bigcup_{g\in S} \mathcal{V}(g)\big)$ for the finite measure $\nu(\cdot) = |\cdot|/A$. Normalisation and monotonicity are immediate. For submodularity, write the marginal gain as the measure of the part of $\mathcal{V}(g)$ not already covered, $f(S\cup\{g\}) - f(S) = \nu(\mathcal{V}(g)\setminus\mathcal{C}(S))$. Since $S \subseteq S'$ implies $\mathcal{C}(S) \subseteq \mathcal{C}(S')$, we get $\mathcal{V}(g)\setminus\mathcal{C}(S') \subseteq \mathcal{V}(g)\setminus\mathcal{C}(S)$ and the claim follows by monotonicity of ν. □

Two quantities drive the whole design. When a guard is *added* to the current set S, the relevant number is its *marginal gain*; when a guard already in S is considered for *removal*, the relevant number is its *exclusive area*.

Definition 5 (Marginal gain and exclusive area)**.** For $g \notin S$ and $g_j \in S$,

$$\delta^+(g \mid S) \;=\; f(S\cup\{g\}) - f(S) \;=\; \nu(\mathcal{V}(g)\setminus\mathcal{C}(S)), \qquad \delta^-(g_j \mid S) \;=\; f(S) - f(S\setminus\{g_j\}).$$

We call $\delta^-(g_j \mid S)$ the *redundancy deficit* of g_j; it is exactly the coverage that would be lost by deleting g_j. A guard with $\delta^- = 0$ is *strictly redundant*.

Proposition 6 (Redundancy is only revealed after saturation)**.** *Let guards be inserted in the order $g_1, g_2, \ldots, g_K$ and write $S_j = \{g_1, \ldots, g_j\}$. Let $\delta_j^+ = \delta^+(g_j \mid S_{j-1})$ be the gain at insertion time, and $\delta_j^- = \delta^-(g_j \mid S_K)$ the redundancy deficit at the end. Then for every j,*

$$\delta_j^- \;\le\; \delta_j^+,$$

and more generally $\delta^-(g_j \mid S)$ is non-increasing in S for $S \ni g_j$.

Proof. Put $B = S_K \setminus \{g_j\}$ and $A' = S_{j-1}$. Then $A' \subseteq B$ and, by proposition 4, $\delta_j^+ = f(A'\cup\{g_j\}) - f(A') \ge f(B\cup\{g_j\}) - f(B) = \delta_j^-$. The monotonicity claim is the same inequality applied to two nested sets. □

Proposition 6 is the formal statement of the intuition behind the staged curriculum requested for this project. A guard that was worth placing can only become *cheaper* to delete as the gallery fills up — never more expensive. Therefore:

Consequence for the curriculum. Deletion decisions taken early in an episode are evaluated against an optimistic (too large) redundancy deficit, and deletion decisions taken after saturation are evaluated against the smallest deficit that guard will ever have. Spending the first stage of training purely on *coverage*, and only then switching on the delete action and the guard cost, is not merely a convenience: it is the regime in which the pruning signal is maximally informative and least noisy.

Two further consequences are worth recording. The greedy policy "repeatedly add the guard of maximum marginal gain" inherits the $1 - 1/e$ guarantee for budgeted coverage [14] and the ln-factor guarantee for full cover [6], which is essentially optimal for set cover; so a well-trained SATURATE stage should approach, not exceed, greedy. The gains available to a learned policy come from (a) choosing *continuous* guard positions rather than a finite candidate set and (b) the pruning that greedy never performs, since greedy solutions are not minimal — a greedy cover can contain strictly redundant elements. This is precisely the slack that PRUNE targets.

3 The decision process

3.1 A contextual family of deterministic MDPs

A single gallery induces a deterministic MDP; the learning problem is over a *distribution* of galleries. We therefore use the contextual MDP formalism [8].

Definition 7 (AGP contextual MDP)**.** Let $\mathcal{D}$ be a distribution over gallery instances. The AGP contextual MDP is

$$\mathfrak{M} \;=\; \langle\, \mathcal{D},\; \mathcal{S},\; \mathcal{A},\; \mathcal{T},\; \mathcal{R},\; \gamma,\; \Omega,\; \mathcal{O} \,\rangle,$$

where for each context $G \sim \mathcal{D}$ the tuple $\langle \mathcal{S}_G, \mathcal{A}, \mathcal{T}_G, \mathcal{R}_G, \gamma \rangle$ is a deterministic finite-horizon MDP, Ω is the observation space and $\mathcal{O} : \mathcal{S} \to \Omega$ the (deterministic, lossy) observation map.

All stochasticity in the environment comes from $\mathcal{D}$ and from the policy; given the instance and the action sequence, the trajectory is determined. This is worth stating because it has a practical consequence: the value function's irreducible variance is entirely *instance* variance, so evaluation must always be reported per difficulty tier, and a fixed held-out instance set is the only meaningful validation signal.

3.2 State

Definition 8 (State)**.** A state at step t is

$$s_t \;=\; (\, G,\; \mathcal{G}_t,\; t,\; \bar{c}_t \,) \;\in\; \mathcal{S},$$

where G is the (fixed) instance, $\mathcal{G}_t = (g_1, \ldots, g_{k_t}) \in \mathcal{F}^{k_t}$ is the ordered tuple of currently placed guards, t is the step index and $\bar{c}_t = \max_{\tau \le t} f(\mathcal{G}_\tau)$ is the running best coverage (used by the coverage ratchet of section 6). We write $k_t = |\mathcal{G}_t|$, $c_t = f(\mathcal{G}_t)$, $u_t = 1 - c_t$.

Remark 9 (Why $\mathcal{G}_t$ and not only $\mathcal{C}_t$)**.** If the only action were Add, the covered region $\mathcal{C}_t$ would be a sufficient statistic: the marginal gain of a new guard depends on the current coverage alone. Once Del is admitted this is false — evaluating a deletion requires knowing *which* guards produced the coverage, i.e. the individual $\mathcal{V}(g_j)$. The guard tuple must therefore be part of the state, and, as section 4 discusses, the observation must carry enough information about it for the policy to act on deletions at all.

3.3 Action space

Definition 10 (Action space)**.**

$$\mathcal{A} \;=\; \underbrace{\{\textsc{Add},\; \textsc{Del},\; \textsc{Stop}\}}_{\text{mode } \mu} \;\times\; \underbrace{[0,1]^2}_{\text{point } a}.$$

Given a state with bounding box $\mathrm{bbox}(\mathcal{F}) = [x_{\min}, x_{\max}] \times [y_{\min}, y_{\max}]$, the normalised point $a = (a_x, a_y)$ is decoded to

$$\xi(a) \;=\; (x_{\min} + a_x(x_{\max} - x_{\min}),\; y_{\min} + a_y(y_{\max} - y_{\min})).$$

The three modes are interpreted as follows.

Add:
place a guard at $\Pi_{\mathcal{F}}(\xi(a))$, where $\Pi_{\mathcal{F}}$ is the projection into the free space defined in eq. (1). A guard is *always* placed.

Del:
delete the guard nearest to the decoded point, $j^\star = \arg\min_{j \le k_t} \|g_j - \xi(a)\|_2$. If $\mathcal{G}_t = \emptyset$ the action is a no-op and incurs the penalty r_{void}.

Stop:
commit the current guard set and terminate the episode.

Remark 11 (Deletion as a geometric pointer)**.** Addressing a guard by a point in the plane rather than by an index keeps the action space fixed-size and permutation-invariant, and reuses the same continuous head as ADD. This mirrors the trick in Xu et al. [22], where the velocity command is not an external input but is *derived from geometry* each step (v_t^{target} is the robot–ball unit vector scaled by a magnitude). Here the same idea turns "which of the k_t guards" — a variable-size discrete choice — into "where", a fixed-size continuous one.

Remark 12 (Backwards compatibility)**.** Masking the mode head to {ADD} recovers exactly the action space of the current implementation, $\mathcal{A} = [0,1]^2$. Stage SATURATE of the curriculum runs under this mask, so no separate environment is needed.

Projection into the free space. The bounding box strictly contains $\mathcal{F}$, so a decoded point may land outside the polygon or inside a hole. Rather than rejecting the action we project:

$$\Pi_{\mathcal{F}}(q) \;=\; \begin{cases} q, & q \in \operatorname{int}\mathcal{F}, \\ q^\star + \theta\,(q_{\text{ref}} - q^\star), & \text{otherwise}, \end{cases} \qquad q^\star = \arg\min_{p \in \mathcal{F}} \|p - q\|_2, \tag{1}$$

where q_{ref} is a fixed interior reference point (Shapely's `representative_point`) and θ is the smallest value in a fixed increasing ladder for which the result is strictly interior. We record the indicator $\iota_t = \mathbf{1}[\xi(a) \notin \operatorname{int}\mathcal{F}]$ for the reward.

This is not cosmetic. Without projection, a deterministic (mean-action) evaluation policy that emits an invalid point places no guard, the observation is unchanged, and the policy re-emits the same invalid point forever — an absorbing deadlock that produced 0% coverage on at least one held-out instance before the fix. Projection guarantees the state advances on every step, which makes the MDP *progressive*: no action can leave the state unchanged. That is a property worth preserving in any future action-space extension.

3.4 Transition

The transition is deterministic. With $\xi = \xi(a)$:

$$\mathcal{G}_{t+1} \;=\; \begin{cases} \mathcal{G}_t \oplus \Pi_{\mathcal{F}}(\xi), & \mu = \text{ADD}, \\ \mathcal{G}_t \ominus g_{j^\star}, & \mu = \text{DEL},\ \mathcal{G}_t \neq \emptyset, \\ \mathcal{G}_t, & \text{otherwise}, \end{cases} \qquad t \mapsto t+1, \qquad \bar{c}_{t+1} = \max(\bar{c}_t,\, f(\mathcal{G}_{t+1})),$$

where $\oplus$ and $\ominus$ are tuple append and removal. Coverage is maintained incrementally: $\mathcal{C}_{t+1} = \mathcal{C}_t \cup \mathcal{V}(g)$ on an add, and $\mathcal{C}_{t+1} = \bigcup_{j \neq j^\star} \mathcal{V}(g_j)$ on a delete (a re-union, which is why the per-guard visibility polygons are cached in `Security.all_guards`).

3.5 Initial state and instance distribution

$s_0 = (G, \emptyset, 0, 0)$ with $G \sim \mathcal{D}$: every episode starts from an *empty* gallery. Document 2 varies $\mathcal{D}$ across stages (difficulty tiers, D_4 symmetry augmentation) and discusses a partially-saturated initialisation as an optional accelerator for PRUNE — the analogue of Wang et al. [21] moving the ball spawn radius from 10 m to 2 m between stages, and of Xu et al. [22] tightening the robot spawn annulus from $[0.5, 4.0]$ m to $[0.5, 2.0]$ m.

3.6 Termination and horizon

Let $c^\dagger$ be the coverage threshold at which an instance counts as solved (currently `COVERAGE_THRESHOLD` $= 0.98$; note the design note text says 0.95, and the two should be reconciled). Let $K_{\max}$ be the guard budget (currently `MAX_GUARDS` $= 20$; section 5.5 recommends 24 to leave room for saturation) and $T_{\max}$ the step budget. An episode ends when

Event	Flag	Condition
Solved	`terminated`	$c_t \geq c^\dagger$ *and* the agent has no pending pruning budget
Committed	`terminated`	$\mu_t = \textsc{Stop}$
Guard budget spent	`truncated`	$k_t \geq K_{\max}$
Step budget spent	`truncated`	$t \geq T_{\max}$
Stalled	`truncated`	ρ consecutive unproductive steps

A step is *unproductive* if $\delta^+ < \eta \cdot u_{t-1}$, i.e. the new guard covers less than a fraction η (`USEFUL_AREA_FRAC` $= 0.10$) of what was still uncovered; the threshold is relative to the remaining gap, so the bar falls as the gallery fills. The patience is $\rho =$ `PATIENCE` $= 3$.

Remark 13 (Solved $\neq$ terminated, once pruning exists)**.** In the add-only setting, terminating at $c_t \geq c^\dagger$ is correct. Once Del exists it is not: the interesting part of the episode — removing the guards that saturation made redundant — happens *after* the threshold is first crossed. In Prune and Refine the episode therefore continues past first coverage and ends on Stop or on the step budget, with Stop carrying a penalty when $c_t < c^\dagger$. This is the reason the Stop action exists at all: it lets the agent choose the episode length, and proposition 17 shows a shorter episode is (weakly) preferred whenever $\gamma < 1$.

Remark 14 (Truncation must bootstrap)**.** All three of the truncation conditions above end an episode at a *non-absorbing* state whose value is genuinely non-zero. Stable-Baselines3 only bootstraps the value at such a state when the environment sets `info["TimeLimit.truncated"] = True` [18]; the current environment does not, so every truncated episode is treated as terminal and its return is systematically underestimated. Combined with a negative per-guard cost this creates a real incentive to stall into the patience truncation. This is a concrete bug to fix alongside the reward redesign.

3.7 Discount

γ is a design parameter, not a property of the problem. Episodes are short (≤ 40 steps), so a discount close to 1 is affordable and desirable: the terminal parsimony payoff must survive back-propagation across the whole episode. We use $\gamma = 0.99$ in Saturate and $\gamma = 0.997$ in Prune/Refine. Note the contrast with the locomotion papers, which run 50 Hz control and thousands of steps per episode and therefore use $\gamma = 0.994$ [21] or $\gamma = 0.995$ [20] — effective horizons of only a few seconds. Those numbers should *not* be copied across; here a step is a guard, not 20 ms.

4 Observation: a POMDP with a privileged critic

The agent does not receive s_t. It receives a rasterised, bounded-size view, which makes the learning problem a POMDP [9]. Following the asymmetric actor–critic pattern [17] used by all three reference papers — Wang et al. [21] give the critic simulator state and a wider first layer (768 vs. 512); Wang et al. [20] and Xu et al. [22] both publish an explicit actor/critic column table — we let the critic see strictly more than the actor.

Table 1 extends the current three-channel image in two ways that the pruning stage requires. The `guards` channel is needed because the union $\mathcal{C}_t$ is indistinguishable between one guard and five guards covering the same region — the existing `stats` vector was added for precisely this reason, but a scalar count cannot tell the policy *where* the guards are, so a Del head has nothing to point at. The `redundancy` channel makes δ_j^- directly legible: a bright guard is expensive to delete, a dark one is nearly free. Without it, the policy would have to infer each guard's exclusive area from the raster, which is exactly the kind of latent quantity that Wang et al. [20] found

Table 1: Observation specification. The actor input is the deployable view; the critic additionally receives privileged quantities that are exact but expensive or non-generalisable. $H = W = 64$ (`IMG_SIZE`).

Component	Shape	Description	Actor	Critic
Raster channels, $H \times W$ each, `uint8`				
`interior`	$H \times W$	$\mathbf{1}[\text{pixel} \in \mathcal{F}]$	✓	✓
`covered`	$H \times W$	$\mathbf{1}[\text{pixel} \in \mathcal{C}_t]$	✓	✓
`remaining`	$H \times W$	$\mathbf{1}[\text{pixel} \in \mathcal{U}_t]$	✓	✓
`guards`	$H \times W$	Gaussian splat of $\mathcal{G}_t$	✓	✓
`redundancy`	$H \times W$	at pixel of g_j: $255 \cdot \min(1, \delta_j^- / \sigma_p)$	✓	✓
`exclusive`	$H \times W$	region seen by exactly one guard		✓
Scalars				
c_t	1	coverage fraction	✓	✓
$k_t / K_{\max}$	1	guard budget used	✓	✓
$t / T_{\max}$	1	step budget used	✓	✓
$\min_j \delta_j^-$	1	cheapest available deletion	✓	✓
$\bar{c}_t - c_t$	1	slack against the coverage ratchet	✓	✓
u_t (exact)	1	exact uncovered fraction (not raster)		✓
k_{ref}	1	greedy reference guard count		✓
difficulty tier	4	one-hot		✓
n, m	2	vertex count, hole count		✓

had to be supplied or explicitly reconstructed — their ablation showed that a decoder trained *after* RL rather than jointly left ball-position predictions "at the noise level".

Design rule (observability of the pruning signal). If the reward depends on δ_j^-, the observation must expose δ_j^-. Otherwise the DEL head is being asked to solve a credit-assignment problem the perception cannot support, and the stage-II gradient will be dominated by noise.

The privileged k_{ref} deserves a note: it is the guard count produced by a greedy max-coverage baseline on the same instance, computed once at dataset generation time. It gives the critic an instance-difficulty normaliser — a hard 40-vertex gallery with 12 holes simply needs more guards than an easy one, and without k_{ref} the value function has to learn that from the raster. It also calibrates the terminal parsimony reward in section 5.5. Note that k_{ref} must stay out of the *actor* observation, or the policy will not generalise to instances with no precomputed baseline.

5 Reward

5.1 The objective we actually want

The AGP objective is terminal and sparse:

$$\max_{\pi} \; \mathbb{E}_{G \sim \mathcal{D},\, \tau \sim \pi} \Big[w_{\text{goal}} \, \mathbf{1}\Big[c_T \geq c^{\dagger} \Big] \; - \; w_{\text{par}} \, k_T \Big]. \tag{2}$$

Everything else in this section exists to make (2) learnable without changing what it prefers.

5.2 Two potentials

Definition 15 (Coverage and parsimony potentials)**.**

$$\Phi_{\text{cov}}(s) \; = \; - \, w_{\text{cov}} \, u(s) \; = \; - \, w_{\text{cov}} \, (1 - f(\mathcal{G}(s))), \qquad \Phi_{\text{par}}(s) \; = \; - \, w_{\text{par}} \, k(s).$$

The associated shaping rewards are $F_\bullet(s, a, s') = \gamma \, \Phi_\bullet(s') - \Phi_\bullet(s)$.

Evaluating F_{cov} and F_{par} at $\gamma = 1$ gives a pleasant surprise: they reproduce exactly the reward terms one would write by hand.

$$F_{\text{cov}} = \begin{cases} +\,w_{\text{cov}}\,\delta^{+}, & \mu = \textsc{Add}, \\ -\,w_{\text{cov}}\,\delta^{-}_{j^\star}, & \mu = \textsc{Del}, \end{cases} \qquad F_{\text{par}} = \begin{cases} -\,w_{\text{par}}, & \mu = \textsc{Add}, \\ +\,w_{\text{par}}, & \mu = \textsc{Del}. \end{cases}$$

So "reward new area, charge for each guard, charge for area lost on deletion, refund the guard cost on deletion" is not four hand-tuned terms — it is two potentials. The requested reward structure ("adding guards is a negative reward, removing them with no space lost is a positive reward, with lower reward as space is lost") falls out of $F_{\text{cov}} + F_{\text{par}}$ with a single free parameter, the ratio $w_{\text{par}}/w_{\text{cov}}$.

Theorem 16 (Policy invariance, Ng et al. [15])**.** *Let $\mathfrak{M}'$ be $\mathfrak{M}$ with reward $r' = r + \gamma\Phi(s') - \Phi(s)$ for any bounded $\Phi : \mathcal{S} \to \mathbb{R}$. If Φ vanishes at absorbing states, then every optimal policy of $\mathfrak{M}'$ is optimal in $\mathfrak{M}$ and vice versa.*

Proposition 17 (The dense reward telescopes into the sparse objective)**.** *Along any trajectory of length T starting from $s_0 = (G, \emptyset, 0, 0)$,*

$$\sum_{t=0}^{T-1} \gamma^t \big(\gamma\Phi(s_{t+1}) - \Phi(s_t)\big) \;=\; \gamma^T \Phi(s_T) - \Phi(s_0).$$

With $\Phi = \Phi_{\text{cov}} + \Phi_{\text{par}}$ and $\gamma = 1$ this equals

$$(-w_{\text{cov}} u_T - w_{\text{par}} k_T) - (-w_{\text{cov}} \cdot 1 - 0) \;=\; w_{\text{cov}}\, c_T \;-\; w_{\text{par}}\, k_T.$$

Hence the undiscounted episode return under the dense reward, plus the terminal bonus, is exactly

$$R(\tau) \;=\; w_{\text{cov}}\, c_T \;-\; w_{\text{par}}\, k_T \;+\; w_{\text{goal}} \mathbf{1}\Big[c_T \geq c^{\dagger}\Big] \;+\; \underbrace{\textstyle\sum_t \gamma^t r_t^{\text{aux}}}_{\textit{auxiliary terms}}.$$

For $\gamma < 1$ the shaping contributes $\gamma^T\Phi(s_T) - \Phi(s_0)$ instead, which additionally discounts the terminal potential by γ^T and thus adds a mild preference for short episodes — a pressure towards fewer guards, and towards using Stop *promptly.*

Proof. The sum telescopes: $\sum_t \gamma^{t+1}\Phi(s_{t+1}) - \sum_t \gamma^t \Phi(s_t) = \gamma^T\Phi(s_T) - \Phi(s_0)$. Substituting $u_0 = 1$, $k_0 = 0$ gives the stated value. □

Proposition 17 is the central design claim of this document. The dense, per-step signal that PPO needs is not an approximation of the sparse AGP objective; it is an exact decomposition of it. This is the same principle Wang et al. [20] rely on when they write that potential-based shaping "guarantees that the learned behaviors remain directed toward successful goal scoring", and it is what lets them give a dense proxy a weight of 500 against a terminal goal reward of 15 without corrupting the objective. It also means we do *not* need to anneal the coverage term away in Prune, unlike Wang et al. [21], who halve their chasing proxy ($2.0 \to 1.0$) precisely because it is *not* potential-based and does bias the optimum.

5.3 Which terms must be annealed

Design rule 18 (Anneal iff non-potential)**.** A reward term may be left at a constant weight for the whole of training if and only if it is expressible as $\gamma\Phi(s') - \Phi(s)$ for a state function Φ, or it is a component of the true objective (2). Every other term biases the optimum and must be scheduled to zero before training ends.

Table 2: Classification of reward terms. Only the last group needs a schedule.

Class	Terms	Treatment
Objective	r_{goal}, r_{pars}, r_{fail}	fixed weights, on from SATURATE
Potential-based	F_{cov}, F_{par}	fixed weights; F_{par} off in SATURATE only because the DEL action is masked
Non-potential	r_{front}, r_{sweep}, r_{sep}	annealed to 0 by the end of SATURATE
Constraint-like	r_{proj}, r_{void}, r_{stop}	fixed; promoted to hard constraints in REFINE

Table 3: The reward basis. One environment, one term list; the curriculum changes only the weight column (Document 2). "Fires" indicates when the term is non-zero. σ_p is the pruning tolerance scale, σ_s the sweep width, $d_{\min}$ the separation radius, ι_t the projection indicator, k_{ref} the greedy reference count.

Symbol	Name	Expression	Fires
Objective group (coverage critic)			
r_{cov}	coverage potential	$\gamma\Phi_{\text{cov}}(s') - \Phi_{\text{cov}}(s)$; equals $+\delta^+$ on ADD, $-\delta^-_{j^\star}$ on DEL (at γ=1)	every step
r_{goal}	coverage attained	$\mathbf{1}\left[c_T \geq c^\dagger\right]$	terminal
r_{fail}	coverage missed	$-\mathbf{1}\left[c_T < c^\dagger\right]$	terminal
Parsimony group (parsimony critic)			
r_{par}	guard-count potential	$\gamma\Phi_{\text{par}}(s') - \Phi_{\text{par}}(s)$; equals -1 on ADD, $+1$ on DEL	every step
r_{prune}	clean-prune bonus	$\exp\!\left(-\delta^-_{j^\star}/\sigma_p\right) \in (0, 1]$	DEL
r_{pars}	terminal parsimony	$\left(k_{\text{ref}} - k_T\right)\mathbf{1}\left[c_T \geq c^\dagger\right]$	terminal
Scaffolding group — annealed to zero (design rule 18)			
r_{front}	frontier bonus	$\mathbf{1}[\Pi_{\mathcal{F}}(\xi) \in \mathcal{U}_t]$	ADD
r_{sweep}	boundary-inward prior	$\exp\!\left(-(d_\partial(\Pi_{\mathcal{F}}(\xi)) - \mu_{\text{ring}}(t))^2/2\sigma_s^2\right)$	ADD
r_{sep}	separation penalty	$-\mathbf{1}[\min_j \lVert\Pi_{\mathcal{F}}(\xi) - g_j\rVert < d_{\min}]$	ADD
Regularity group — constraints in REFINE			
r_{proj}	aimed outside $\mathcal{F}$	$-\iota_t$	ADD
r_{void}	deleted from empty set	$-\mathbf{1}[\mathcal{G}_t = \emptyset]$	DEL
r_{stop}	premature commit	$-\mathbf{1}\left[c_t < c^\dagger\right]$	STOP

This gives an objective criterion for something the reference papers do by hand. In Wang et al. [21], the reference-motion imitation reward is dropped $1.2 \to 0.0$ with the stated reason that it is "removed to promote exploration and discovery of effective dribbling behaviors"; in Xu et al. [22], feet-swing ($3.0 \to 0.0$), arm-swing ($1.0 \to 0.0$) and feet-yaw ($-1.0 \to 0.0$) are all zeroed at the stage boundary. All of these are non-potential shaping terms and all of them are annealed; the terms those papers keep alive are the ones closest to the task objective. Design rule 18 says the pattern is not a coincidence.

Under this rule our terms split as in table 2.

5.4 The reward basis

Table 3 lists every term. Three remarks on the less obvious entries.

r_{prune} and the "lower reward as space is lost" requirement. The bounded exponential kernel $\exp(-\delta^-/\sigma_p)$ equals 1 for a strictly redundant guard and decays smoothly to 0 as the

deletion becomes expensive. Bounding each shaping term in $(0, 1]$ is the single most consistent stylistic choice across the reference papers — Wang et al. [21] write *every* tracking reward as $\exp\{-\alpha\|x - x^\star\|^2\}$, and Xu et al. [22] use $\exp(-(v^{\text{target}} - v)^2/\sigma)$. The practical payoff is that a weight then *is* the term's maximum per-step contribution, which makes hand-balancing a dozen terms tractable. Note that r_{prune} is technically non-potential and therefore does bias the optimum; it survives design rule 18 only because the bias is negligible ($\sigma_p \ll 1$, so it is effectively a delta function at $\delta^- \approx 0$, i.e. a tie-breaker among genuinely free deletions). If purity is preferred, it can be annealed with the scaffolding group.

r_{sweep} and the "edges first, then inwards" idea. The ring target

$$\mu_{\text{ring}}(t) \;=\; \mu_0 + (\mu_1 - \mu_0)\,\frac{t}{K_{\max}}, \qquad 0 \le \mu_0 < \mu_1, \tag{3}$$

sweeps the preferred distance-to-boundary outward-to-inward *within* an episode: early guards are rewarded for sitting near $\partial\mathcal{F}$ (where reflex vertices and hole boundaries create the occlusions), later ones for moving into the interior. This is a prior on the *order* of placement, not on the final configuration, and it is the AGP analogue of the gait-phase indicators (`feet swing` × `swing period`) that Xu et al. [22] use to install a walking rhythm and then delete. Both μ_0, μ_1 should be expressed as fractions of the bounding-box diagonal so the term is scale-free. An alternative formulation replaces d_∂ by distance to the nearest reflex vertex, $\min_{v\in\mathcal{R}}\|\cdot - v\|$, which is better motivated by the geometry (reflex vertices are exactly the sources of occlusion, and are what makes $\lfloor n/3 \rfloor$ tight [3, 16]) but requires computing $\mathcal{R}$ per instance.

r_{pars} and calibration against a reference solver. Rewarding $-k_T$ directly makes the terminal reward's scale instance-dependent: an easy gallery needs 3 guards, an extreme one 15, so the same $-k_T$ means very different things. Measuring against the greedy baseline, $(k_{\text{ref}} - k_T)$, normalises this and gives a directly interpretable quantity — "guards saved relative to greedy". This mirrors Xu et al. [22], who set their constraint threshold to "1.5× per-step average regularization cost of a walking gait", i.e. calibrated against a known-good reference behaviour rather than in absolute units.

5.5 Weight calibration

Three inequalities determine the weights; everything else is tuning.

(C1) The guard–area exchange rate. The ratio

$$\epsilon \;=\; \frac{w_{\text{par}}}{w_{\text{cov}}} \tag{4}$$

is the fraction of gallery area that one guard is worth. It is the single most interpretable hyperparameter in the whole design, and it should be set from the problem, not by sweeping: if a typical solved instance uses k_{typ} guards, then $\epsilon \approx 1/(2k_{\text{typ}})$ says "a guard is worth half of its fair share of the gallery", which keeps coverage dominant while still making redundancy expensive. With $k_{\text{typ}} \approx 8$–12 across the four tiers we take $\epsilon = 0.02$, i.e. $w_{\text{cov}} = 1.0$, $w_{\text{par}} = 0.02$.

(C2) Feasibility must dominate parsimony. The agent must never prefer a smaller infeasible solution to a larger feasible one. Since at most $K_{\max}$ guards can be placed, it suffices that

$$w_{\text{goal}} \;+\; w_{\text{fail}} \;>\; w_{\text{par}}\,K_{\max} \;+\; w_{\text{pars}}\,k_{\text{ref}}^{\max}. \tag{5}$$

With $w_{\text{par}} = 0.02$, $K_{\max} = 24$, $w_{\text{pars}} = 0.25$ and $k_{\text{ref}}^{\max} \approx 18$ the right-hand side is ≈ 4.98, so $w_{\text{goal}} = 4.0$ and $w_{\text{fail}} = 2.0$ satisfy (5) with margin. This is the AGP analogue of Wang et al. [20] setting their fall penalty to -1000 against task rewards of order 10–500: the hard-failure term is deliberately off-scale.

(C3) The pruning tolerance. A deletion is immediately profitable when

$$w_{\text{par}} + w_{\text{prune}} \exp\!\big(-\delta^{-}/\sigma_p\big) \;>\; w_{\text{cov}}\, \delta^{-}.$$

The left side is decreasing in δ^{-} and the right side increasing, so there is a unique root $\epsilon^{\star}$ — the *pruning tolerance*, the largest coverage loss the agent will accept in exchange for one fewer guard:

$$\epsilon^{\star} \;=\; \frac{w_{\text{par}}}{w_{\text{cov}}} + \frac{w_{\text{prune}}}{w_{\text{cov}}}\, e^{-\epsilon^{\star}/\sigma_p} \;\approx\; \epsilon \;+\; \frac{w_{\text{prune}}}{w_{\text{cov}}}\, e^{-\epsilon/\sigma_p} \quad \text{when } \sigma_p \ll \epsilon. \tag{6}$$

With $w_{\text{cov}} = 1.0$, $w_{\text{par}} = 0.02$, $w_{\text{prune}} = 0.15$, $\sigma_p = 0.005$, the root is $\epsilon^{\star} \approx 0.022$: the agent will trade away up to about 2.2% of the gallery to remove a guard, and essentially nothing beyond that. Note that $\epsilon^{\star}$ is dominated by the exchange rate ϵ, which is the intended behaviour — the exponential bonus only breaks ties among nearly-free deletions. The hard feasibility floor is enforced separately by the ratchet of section 6, so $\epsilon^{\star}$ governs *intermediate* states, not the final answer.

Default values.

Symbol	Value	Set by
w_{cov}	1.0	normalisation (defines the unit of reward)
w_{par}	0.02	(C1), exchange rate $\epsilon = 1/(2k_{\text{typ}})$
w_{goal}	4.0	(C2)
w_{fail}	2.0	(C2)
w_{pars}	0.25	(C2), scale of "guards saved vs. greedy"
w_{prune}	0.15	(C3)
σ_p	0.005	(C3), 0.5% of gallery area counts as "no space lost"
$\epsilon^{\star}$	0.022	derived from (C1)+(C3)
$c^{\dagger}$	0.98	task specification
$K_{\max}$	24	budget; must exceed $k_{\text{ref}}^{\max}$ with slack for saturation

5.6 Reward groups and the multi-critic split

The rows of table 3 are grouped so that each group can be given its own value head, following Wang et al. [20], who report that "using a single critic to estimate combined rewards can cause negative interference between distinct reward components, leading to reduced learning stability and performance" and combine advantages as $A_{\text{total}} = w_{\text{goal}} A_{\text{goal}} + w_{\text{aux}} A_{\text{aux}}$ with weights 2 and 1. Our split is

$$A_{\text{total}} \;=\; \beta_{\text{cov}}\, A_{\text{cov}} \;+\; \beta_{\text{par}}\, A_{\text{par}} \;+\; \beta_{\text{reg}}\, A_{\text{reg}}, \tag{7}$$

with the coverage group, the parsimony group, and the scaffolding-plus-regularity group respectively. The reason this matters here is scale: r_{cov} has per-step magnitude up to ~ 0.5 early in an episode and $\sim 10^{-3}$ late, while r_{par} is a constant ± 0.02. A single critic must fit both regimes with one output head; two heads need not. The stage-dependent β schedule is set in Document 2.

6 Constraints

Two requirements are better expressed as constraints than as reward terms, giving a constrained MDP [2].

Validity. $g \in \mathcal{F}$ for every placed guard. This is enforced structurally by $\Pi_{\mathcal{F}}$ (eq. (1)), with r_{proj} retained as a soft penalty so the policy still learns to aim inside rather than relying on the projection.

The coverage ratchet. Once the agent has achieved coverage $\bar{c}_t$ it should not be allowed to fall back arbitrarily far while pruning:

$$c_t \;\geq\; \min(\bar{c}_t,\; c^{\dagger}) - \delta_{\text{slack}}, \qquad \delta_{\text{slack}} = 0.01. \tag{8}$$

In PRUNE this is implemented as an *action mask*: a DEL whose $\delta^{-}_{j^\star}$ would violate (8) is rejected and charged r_{void}. In REFINE it is promoted to a state-wise constraint solved with a constrained policy-gradient method:

$$\max_{\pi}\; \mathbb{E}\Big[\textstyle\sum_t \gamma^t r^{\text{task}}_t\Big] \quad \text{s.t.} \quad c^{\text{cost}}(s_t, a_t) \leq h \quad \forall (s_t, a_t) \in \tau,\; \tau \sim p_\pi. \tag{9}$$

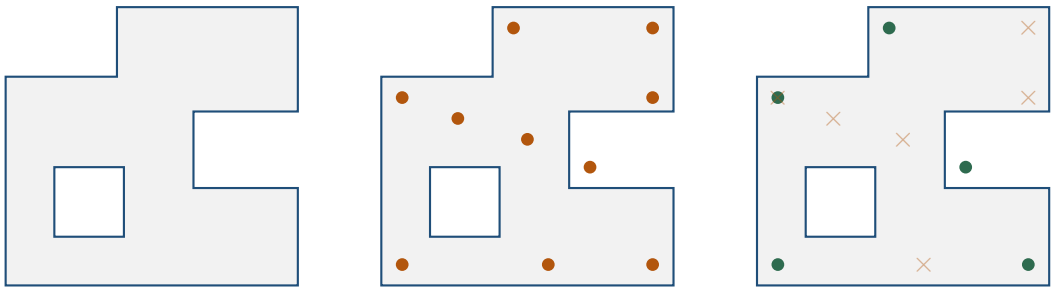
Equation (9) is the formulation Xu et al. [22] adopt (their Eqs. 4–5, solved with N-P3O [12]), and their motivation transfers directly. They diagnose *inhomogeneous credit assignment*: task reward is concentrated at one moment of the episode (the kick), so "a fixed regularization coefficient can impose disproportionately large penalties at certain steps — particularly those immediately preceding the kick, where the agent expects high immediate future reward". Their ablation is the strongest part of the argument: raising the fixed coefficient from 1.0 to 1.5 *lowered* success from 64.8% to 43.4%, while the constrained formulation reached 79.5%. AGP has exactly the same profile — the first few guards each capture a large share of the area and the last ones capture slivers, so any fixed per-step penalty is mis-scaled at one end of the episode or the other.

7 The environment step, in full

Algorithm 1 One environment step of $\mathfrak{M}$

Require: state $s_t = (G, \mathcal{G}_t, t, \bar{c}_t)$, action (μ, a), stage weight vector w
$\xi \leftarrow \xi(a)$ ▷ decode into gallery coordinates
if $\mu =$ STOP **then**
  return s_t, $r = w_{\text{stop}}\, r_{\text{stop}} +$ TERMINAL(s_t), `terminated`
else if $\mu =$ ADD **then**
  $\iota \leftarrow \mathbf{1}[\xi \notin \operatorname{int} \mathcal{F}]$; $g \leftarrow \Pi_{\mathcal{F}}(\xi)$
  $\delta^{+} \leftarrow \nu(\mathcal{V}(g) \setminus \mathcal{C}_t)$ ▷ exact, via Shapely difference
  $\mathcal{G}_{t+1} \leftarrow \mathcal{G}_t \oplus g$; cache $\mathcal{V}(g)$
  $r \leftarrow w_{\text{cov}}\delta^{+} - w_{\text{par}} + w_{\text{front}} r_{\text{front}} + w_{\text{sweep}} r_{\text{sweep}} + w_{\text{sep}} r_{\text{sep}} - w_{\text{proj}}\iota$
else ▷ $\mu =$ DEL
  if $\mathcal{G}_t = \emptyset$ **then**
    return s_t, $r = -w_{\text{void}}$, continue
  end if
  $j^\star \leftarrow \arg\min_j \|g_j - \xi\|_2$; $\delta^{-} \leftarrow f(\mathcal{G}_t) - f(\mathcal{G}_t \ominus g_{j^\star})$
  if $f(\mathcal{G}_t) - \delta^{-} < \min(\bar{c}_t, c^{\dagger}) - \delta_{\text{slack}}$ **then**
    return s_t, $r = -w_{\text{void}}$, continue ▷ ratchet violation, masked
  end if
  $\mathcal{G}_{t+1} \leftarrow \mathcal{G}_t \ominus g_{j^\star}$
  $r \leftarrow -w_{\text{cov}}\delta^{-} + w_{\text{par}} + w_{\text{prune}} \exp(-\delta^{-}/\sigma_p)$
end if
$c_{t+1} \leftarrow f(\mathcal{G}_{t+1})$; $\bar{c}_{t+1} \leftarrow \max(\bar{c}_t, c_{t+1})$
evaluate termination (section 3.6); on terminal add TERMINAL(s_{t+1})
on truncation: set `info["TimeLimit.truncated"] = True` ▷ remark 14
return s_{t+1}, r, flags

function TERMINAL(s)
  return $w_{\text{goal}}\mathbf{1}[c \geq c^{\dagger}] - w_{\text{fail}}\mathbf{1}[c < c^{\dagger}] + w_{\text{pars}}(k_{\text{ref}} - k)\mathbf{1}[c \geq c^{\dagger}]$
end function



(a) s_0: empty gallery (b) SATURATE: saturate, $k = 10$ (c) PRUNE: prune, $k = 5$

Figure 1: The saturate-then-prune trajectory that the MDP is built to support. Proposition 6 says each guard's redundancy deficit δ_j^- is smallest in panel (b), so that is the state in which deletions (crosses in panel (c)) are cheapest and the pruning signal is least ambiguous.

8 Mapping onto the current implementation

Table 4 places every reward term currently in `ppo_agp.py` into the formalism above.

Table 4: The existing reward, classified. "Invariant" means the term does not change the optimal policy and may therefore keep a fixed weight (design rule 18).

Code	Formal term	Invariant	Comment
`area_term`	F_{cov} on ADD	yes	already exactly δ^+
`cost_term`	F_{par} on ADD	yes	$w_{\text{par}} =$ `GUARD_COST` $= 0.05$, i.e. $\epsilon = 0.05$ — generous; (C1) suggests 0.02
`placement_term`	r_{front}	no	biases towards placing in uncovered spots even when a covered spot would see more; anneal
`spacing_term`	r_{sep}	no	a fixed 15%-of-diagonal exclusion is wrong for corridors; anneal
`invalid_term`	r_{proj}	(constraint)	keep
—	r_{goal}, r_{fail}, r_{pars}	—	**absent**: no terminal reward of any kind
—	r_{prune}, DEL	—	**absent**: the action space has no removal

Four gaps follow, in rough order of impact.

G1. No terminal reward. Proposition 17 shows the potential terms already encode $w_{\text{cov}} c_T - w_{\text{par}} k_T$, but there is no $\mathbf{1}\left[c_T \geq c^\dagger\right]$ term, so crossing the threshold is worth nothing beyond the area itself. Combined with **G2** this makes giving up cheap.

G2. Truncation is treated as termination (remark 14). The patience stall, the guard-budget exhaustion and the step limit all cut the episode at a state with real remaining value, and none of them sets `TimeLimit.truncated`. The value target is therefore biased low exactly on the trajectories the agent should learn to avoid. This is a plausible contributor to the training instability recorded in the design note (approx_kl in the tens, clip_fraction ≈ 0.8, collapse after 150–200k steps) — not the only one, but a cheap fix to rule out.

G3. No removal action, so the pruning stage cannot be expressed. This is the action-space extension of definition 10.

G4. The observation cannot support pruning even if the action existed: guards are not visible as individual objects and δ_j^- is not exposed (table 1).

A fifth, smaller item: `COVERAGE_THRESHOLD` is 0.98 in code and 0.95 in the prose of `docs/agp_ppo_design.md`; the calibration in section 5.5 assumes 0.98.

9 Evaluation metrics

The MDP return is a training signal, not a report. Evaluation on the held-out set should report, per difficulty tier:

- **Feasibility rate** $\mathbb{P}[c_T \geq c^\dagger]$ — the primary number.
- **Mean coverage** $\mathbb{E}[c_T]$, including failures.
- **Guard count** $\mathbb{E}[k_T \mid c_T \geq c^\dagger]$, conditioned on success so that failures cannot flatter the count.
- **Competitive ratio** $\mathbb{E}[k_T / k_{\text{ref}}]$ against greedy, and against an exact IP solution [4, 19] on the `easy` tier where one is computable.
- **Pruning yield** $\mathbb{E}[k_{\text{peak}} - k_T]$: how many guards PRUNE removes. This is the quantity the whole second stage exists to increase, and it is zero by construction for any add-only policy.
- **Redundancy at termination** $\mathbb{E}[\#\{j : \delta_j^- < \sigma_p\}]$: strictly redundant guards left behind. A well-pruned solution should have none.

Reporting k_T without conditioning on feasibility is the easiest way to be fooled here: a policy that stops after two guards has an excellent guard count.

10 Summary

1. The AGP is modelled as a deterministic contextual MDP over gallery instances, with state $(G, \mathcal{G}_t, t, \bar{c}_t)$ and action space $\{\text{ADD}, \text{DEL}, \text{STOP}\} \times [0,1]^2$, where DEL addresses a guard geometrically rather than by index.
2. Coverage is a monotone submodular set function, so a guard's redundancy deficit can only shrink as the gallery fills (proposition 6). Saturating first and pruning second is therefore the ordering in which the pruning signal is strongest.
3. The dense reward consists of exactly two potentials, on uncovered area and on guard count. Their discounted sum telescopes into the true sparse objective (proposition 17), so the dense signal is unbiased by construction and the curriculum can reweight freely.
4. A term may keep a fixed weight iff it is potential-based or part of the objective (design rule 18); this yields an explicit, principled list of which terms the curriculum must anneal to zero, rather than the hand-chosen schedules used in the reference papers.
5. Weights are fixed by three inequalities: the guard–area exchange rate $\epsilon = w_{\text{par}}/w_{\text{cov}}$, feasibility dominance, and the pruning tolerance $\epsilon^\star$, which comes out at about 2.2% of gallery area for the defaults.
6. The current implementation is the ADD-only special case with no terminal reward, two biasing shaping terms, and truncation handled as termination. All four gaps are addressable without changing the environment's external interface.

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